Subiecte 2013 - 14 ianuarie. Prog Radu

1) Spline C^2

Tensionat

$$A = \begin{bmatrix} 2h_0 & h_0 & 0 & \cdots & 0 \\ h_0 & 2(h_0+h_1) & h_1 & & \\ 0 & h_1 & 2(h_1+h_2) & h_2 & \\ & & \ddots & \ddots & \ddots \\ & & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} \\ 0 & & \cdots & h_{n-1} & 2\cdot h_{n-1} \end{bmatrix} \quad x = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix}$$

$$b = \begin{bmatrix} \frac{3}{h_0}(a_1 - a_0) - 3f'(a) \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ \frac{3}{h_{n-1}}(a_n - a_{n-1}) - \frac{3}{h_{n-2}}(a_{n-1} - a_{n-2}) \\ 3f'(b) - \frac{3}{h_{n-1}}(a_n - a_{n-1}) \end{bmatrix}$$

Nat	Tens
$S''(x_0) = 0$	$S'(x_0) = f'(x_0)$
$S''(x_n) = 0$	$S'_{n-1}(x_n) = f'(x_n)$

2) Spline tensionat C^2

$f(x)$: $x = \{0, 1, 2\}$; $f = \{6, 4, 3\}$; $f' = \{1, x, -1\}$;

$$\begin{cases} S_0(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 \\ S_1(x) = a_1 + b_1(x - x_1) + a_1(x - x_1)^2 + d_1(x - x_1)^3 \end{cases}$$

$$\begin{cases} S_i(x_{i+1}) = S_{i+1}(x_{i+1}) \\ S'_i(x_{i+1}) = S'_{i+1}(x_{i+1}) \\ S''_i(x_{i+1}) = S''_{i+1}(x_{i+1}) \\ S_i(x_i) = f(x_i) \\ S_{n-1}(x_n) = f(x_n) \end{cases} \Rightarrow \begin{cases} b_0 + 2c_0(x - x_0) + 3d_0(x - x_0)^2 = S_0'(x) \\ b_1 + 2c_1(x - x_1) + 3d_0(x - x_1)^2 = S_1'(x) \\ 2c_0 + 6d_0(x - x_0) = S_0''(x) \\ 2c_1 + 6d_1(x - x_1) = S_1''(x) \end{cases}$$

$S_0(x_0) = f(x_0) \Rightarrow a_0 = 6.$

$S_1(x_1) = f(x_1) \Rightarrow a_1 = 4$

$S_1(x_2) = f(x_2) = 3$

$\Leftrightarrow 4 + b_1 \cdot 1 + c_1 + d_1 \cdot 1 = 3$

$\Rightarrow \underline{b_1 + c_1 + d_1 = -1} \quad (1)$

	a_i	b_i	c_i	d_i
0	6	1		
1	4.			

$S_0(x_1) = S_1(x_1)$

$6 + b_0 \cdot 1 + c_0 \cdot 1 + d_0 \cdot 1 = 4 \Rightarrow b_0 + c_0 + d_0 = -2. \quad (2)$

$S_0'(x_1) = S_1'(x_1)$

$b_0 + 2c_0 + 3d_0 = b_1 \Rightarrow b_0 - b_1 + 2c_0 + 3d_0 = 0. \quad (3)$

$S_0''(x_1) = S_1''(x_1)$

$\Rightarrow 2c_0 + 6d_0 = 2c_1 \Rightarrow c_0 - c_1 + 3d_0 = 0. \quad (4)$

$S_0'(x_0) = f'(x_0) = 1$

$b_0 = 1.$

$S_1'(x_2) = f'(x_2) = -1$ ~~$b_1 = 1$~~

$b_1 + 2c_1 + 3d_1 = -1 \quad (5)$

$$\begin{cases} b_1 + c_1 + d_1 = -1 \\ c_0 + d_0 = -3 \\ -b_1 + 2c_0 + 3d_0 = 0 \\ c_0 - c_1 + 3d_0 = 0 \\ b_1 + 2c_1 + 3d_1 = -1. \end{cases}$$

4) Cum pot fi îmbunătățite formulele Gaussiene în comparație cu Newton-Cotes

Newton Cotes fol. pct echidistante.

Integrarea Gaussiană fol. polin ortogonale.

O cuadratură Gaussiană de n pct aprox exact polin de grad $2n-1$ sau mai mic

5) Alg. GRAM + conj Fourier!

$$\begin{cases} \langle u_1, u_1 \rangle c_1 + \langle u_2, u_1 \rangle c_2 + \dots + \langle u_{n-1}, u_1 \rangle c_{n-1} = \langle f, u_1 \rangle \\ \langle u_1, u_2 \rangle c_1 + \langle u_2, u_2 \rangle c_2 + \dots + \langle u_{n-1}, u_2 \rangle c_{n-1} = \langle f, u_2 \rangle \\ \vdots \\ \langle u_1, u_n \rangle c_1 + \langle u_2, u_n \rangle c_2 + \dots + \langle u_n, u_n \rangle c_n = \langle f, u_n \rangle \end{cases}$$

Sistem GRAM -> simetric și rău cond. al cărui det = det GRAM, îl rezolvă greu așa că se încearcă diferită pt o bază ortog.
=> sistem diagonal, iar $c_n^* = \langle f, u_n \rangle$ = coef Fourier.

7) Scrieți o fct care impl Met Euler, incluzând pt fct f dată ca parcurs.

$\frac{dy}{dx} = f(x,y)$
$y_{i+1} = y_i + f(x_i, y_i) * h$

```
function y = euler(a,b,m,f,y0)
    x = linspace(a,b,m)
    h = x(2) - x(1);
    y(1) = y0;
    for i = 1:m
        y(i+1) = y(i) + f(x(i), y(i)) * h;
    endfor
endfunction
function rez = f(x,y)
    rez = 2*x + y;
endfunction
```

Apel:
euler(0,3,10,@f,0.5)

8) Calc. aprox $\int_0^1 \frac{x^4}{\sqrt{x(1-x)}}\,dx$ folosind cuad cu $\int_{-1}^{1} \frac{f(x)}{\sqrt{(1-x)^2}}\,dx$

Facem sch de var $t = 2x-1 \Rightarrow x = \frac{t+1}{2}$
$dt = 2dx \quad dx = \frac{dt}{2}$

$$\int_0^1 \frac{x^4}{\sqrt{x(1-x)}} = \int_{-1}^{1} \frac{\left(\frac{t+1}{2}\right)^4}{\sqrt{\left(\frac{t+1}{2}\right)\left(1-\frac{t+1}{2}\right)}} \cdot \frac{dt}{2} = \frac{1}{2}\cdot\int_{-1}^{1} \frac{\left(\frac{1}{2}\right)^4 \cdot (t+1)^4}{\sqrt{\frac{1}{2}(t+1)\left(\frac{1}{2}-\frac{1}{2}t\right)}}\,dt.$$

$$= \frac{1}{2}\cdot\int_{-1}^{1} \frac{\left(\frac{1}{2}\right)^4 \cdot (t+1)^4}{\frac{1}{2}\sqrt{(1+t)(1-t)}}\,dt = \frac{1}{2^4}\int_{-1}^{1} \frac{(t+1)^4}{\sqrt{1-t^2}}\,dt \longrightarrow f(t)$$

2014 -> 9 iunie

1) Explicați conceptul în sensul CMMP

Se încearcă o linie/ polinom care să treacă printre pct date și să aprox fct cât mai bine - Ne-am dori ca eroarea să fie minimă

$$E_2(a_0, a_1) = \sum_{i=1}^{m} [(y_i - (a_1 x_i + a_0)]^2$$

2) Scrieți o fct Matlab/Octave care impl. polin Lagrange.

```
U = diag(x)*ones(n) - ones(n)*diag(x) + eye(n);
V = a - diag(x)*ones(n) + eye(n)
l = [l1 l2 ... ln] = prod(V)./prod(U)
```

Valoarea în $a = b = \sum_{i=1}^{n} l_i \cdot y_i$ = prod(V)./prod(U)*y;

3) FFT calc. Direct Fourier Transf sau inv. Fourier Transf de ord unei frecvențe. Grupează termeni într-un produs a.î reduce complexitatea de la $O(n^2)$ la $O(n \log(n))$

4) Integrare Gaussiană

Newton-Cotes cu grad de val. cel mult n, integrarea Gaussiană extinde valabilitatea la $2n+1$.

Se scrie sub forma de cuadraturi

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} c_i \cdot f(x_i)$$

5/ Det. cuadraturea pt $\int_{-1}^{1} f(x)dx \simeq a_1 f(x_1) + a_2 f(x_2)$ a.i formula de integrare nu are grad max de val.

$$\int_{-1}^{1} f(x)dx \simeq a_1 f(x_1) + a_2 f(x_2)$$

$$\int_a^b g(x) w(x) dx \simeq \sum_{i=1}^{N} A_i f(x_i)$$

$$g(x) = \prod_{i=1}^{n} (x - x_i)$$

$w(x) = 1$, $[a,b] = [-1,1]$

$g(x) = (x-x_1)(x-x_2) = x^2 + ax + b$

$$\int_{-1}^{1} g(x) \cdot w(x) \cdot x^k dx = 0, \quad k = \overline{0,1}$$

I/ $k = 0$.

$$\int_{-1}^{1} x^2 + ax + b\, dx = 0 \Rightarrow \left(\frac{x^3}{3} + a \cdot \frac{x^2}{2} + bx \right) \Big|_{-1}^{1} = 0$$

$$\frac{1}{3} + a \cdot \frac{1}{2} + b - \frac{(-1)}{3} - \frac{a}{2} - \left(\frac{-b}{3}\right) = 0$$

$$\Rightarrow b = -\frac{1}{3}$$

II/ $k = 1$.

$$\int_{-1}^{1} x^3 + ax^2 + bx\, dx = 0$$

$$\left(\frac{x^4}{4} + a \cdot \frac{x^3}{3} + b\frac{x^2}{2} \right) \Big|_{-1}^{1} = 0$$

$$\frac{1}{4} + a \cdot \frac{1}{3} + b \cdot \frac{1}{2} - \frac{1}{4} - \frac{(-1)}{3} \cdot a - \frac{1}{2} \cdot b = 0$$

$$\frac{2}{3} \cdot a = 0 \Rightarrow \boxed{a = 0}$$

$$g(x) = x^2 - \frac{1}{3} = 0 \Rightarrow \begin{cases} x_1 = \sqrt{\frac{1}{3}} \\ x_2 = -\sqrt{\frac{1}{3}} \end{cases}$$

Ca să aflăm pe a_1, a_2 integrala trb să fie exactă pt

(I) $f(x) = 1 \Rightarrow \int_{-1}^{1} 1\, dx = \boxed{a_1 + a_2 = 2}$

(II) $f(x) = x \Rightarrow \int_{-1}^{1} x\, dx = \frac{x^2}{2}\Big|_{-1}^{1} = \frac{1}{2} - \frac{1}{2} = 0 = a_1 \cdot \underbrace{\sqrt{\frac{1}{3}}}_{x_1} + a_2 \underbrace{\left(-\sqrt{\frac{1}{3}}\right)}_{x_2} \Rightarrow$

$$\Rightarrow \begin{cases} a_1 = 1 \\ a_2 = 1 \end{cases}$$

G.V $= 2 \cdot 2 - 1 = 3 \Rightarrow \int_{-1}^{1} f(x)dx \simeq 1 \cdot f\left(\sqrt{\frac{1}{3}}\right) + 1 \cdot f\left(-\sqrt{\frac{1}{3}}\right)$

$G_2 \cdot B_1 = \begin{pmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & 0 & 1 \end{pmatrix}$.

b) Ce înseamnă aprox continuă tigon în sensul o.p.m.p?

$S_n(x) = \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cdot \cos(kx) + b_k \cdot \sin(kx))$

$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \cos(kx)\, dx$

$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \sin(kx)\, dx$

$f(x) = \begin{cases} 0, & -\pi < x \le 0 \\ 1, & 0 < x < \pi \end{cases}$

$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx = \frac{1}{\pi} \int_{-\pi}^{0} 0\, dx + \frac{1}{\pi} \int_{0}^{\pi} 1\, dx = 1$

$a_k = \frac{1}{\pi} \cdot \int_{-\pi}^{\pi} f(x) \cdot \cos(kx)\, dx$

$= \frac{1}{\pi} \cdot \int_{-\pi}^{0} 0 \cdot \cos(kx)\, dx + \frac{1}{\pi} \int_{0}^{\pi} 1 \cdot \cos(kx)\, dx$

$= \frac{1}{\pi} \cdot \frac{\sin(kx)}{k} \Big|_0^{\pi} = 0.$

$b_k = \frac{1}{\pi} \cdot \int_{-\pi}^{\pi} f(x) \cdot \sin(kx)\, dx = \frac{1}{\pi} \int_{-\pi}^{0} 0 \cdot \sin(kx)\, dx + \frac{1}{\pi} \int_{0}^{\pi} 1 \cdot \sin(kx)\, dx$

$= -\frac{1}{\pi} \cdot \frac{\cos kx}{k} \Big|_0^{\pi} = -\frac{1}{\pi} \cdot \left(\frac{(-1)^k - 1}{k} \right)$

$= \frac{1}{\pi} \cdot \left(\frac{1 - (-1)^k}{k} \right)$

$S_n(x) = \frac{1}{2} + \frac{1}{\pi} \cdot \sum_{k=1}^{n} \left(\left(\frac{1 - (-1)^k}{k} \right) \cdot \sin(kx) \right)$

8) Aplicații Householder (un pas pe matrice)

$A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}$; $A^{(1)} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$; $\det(\lambda I - A^{(1)}) = 0$

$\begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (\lambda - 4)(\lambda - 2)$

$\Rightarrow \lambda_1 = 2$

$\lambda_2 = 4$

Alegem $\nabla_1 = 2$

$B_1 = A^{(1)} - \nabla_1 I_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

$G_2 = \begin{pmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$\rho = \sqrt{2}$

$s = \frac{1}{\sqrt{2}}$; $c = \frac{-1}{\sqrt{2}}$

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$G_{12} \cdot B_1 = \begin{pmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \end{pmatrix}$;

$c = g_{ll}$ $g_{lk} = -b$ $\begin{pmatrix} c & -b \\ b & c \end{pmatrix}$

$c = g_{kk}$ $g_{kl} = b$

$G_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & b \\ 0 & b & c \end{pmatrix}$ $\quad s = \sqrt{A(l,l)^2 + A(k,l)^2}$

$\cos = \frac{A(l,l)}{s}$; $\sin = -\frac{A(k,l)}{s}$

$G_{23} \cdot G_{12} \cdot B_1 = \begin{pmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \\ 0 & 0 & \frac{-1}{\sqrt{2}} \end{pmatrix}$

$\underbrace{G_{23} \cdot G_{12} \cdot B_1}_{Q^{(1)t}} = R^{(1)} \Rightarrow Q^{(1)} = G_{12}^t \cdot G_{23}^t = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{-1}{\sqrt{2}} \\ 0 & 1 & 0 \end{pmatrix}$

$A^{(2)} = R^{(1)} \cdot Q^{(1)} + \nabla_1 I_3$

$A^{(2)} = \begin{pmatrix} 2 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 1 & -\frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & 0 \end{pmatrix} + \nabla_1 I_3 \Rightarrow$ încă un pas.

$\neq 0$

Var. A - 2013

1) La ce se referă clasa unui Spline?

Grad de continuitate al fct Spline.

Ce clasă au Spline-urile ce folosesc fct de îmb polin de grad 3? C_1 sau C_2.

Explicați: tot ce știm despre Splines-uri, proprietăți

$S_i(x) = a_0^i + b_0^i(x - x_0) + c_0^i(x - x_0)^2 + d_0^i(x - x_0)^3$

Pt clasa C_2:

$$\begin{cases} S_i(x_i) = f(x_i) ; S_i(x_{i+1}) = S_{i+1}(x_i) \\ S'_{i+1}(x_i) = S'_i(x_i) \\ S''_{i+1}(x_i) = S''_{i+1}(x_i) \end{cases}$$

naturale	funcționale
$S''(x_0) = 0$	$S'(x_0) = f'(x_0)$
$S''(x_n) = 0$	$S'(x_n) = f'(x_n)$

2/ Arăt. că g^* e unic!

$\langle f - g^*, g \rangle = 0$, $\forall g \in G_1$, g^* cel mai bun aprox al lui f

Presp. că (3) g_1^* și g_2^* cei mai buni aproximanți ai lui f

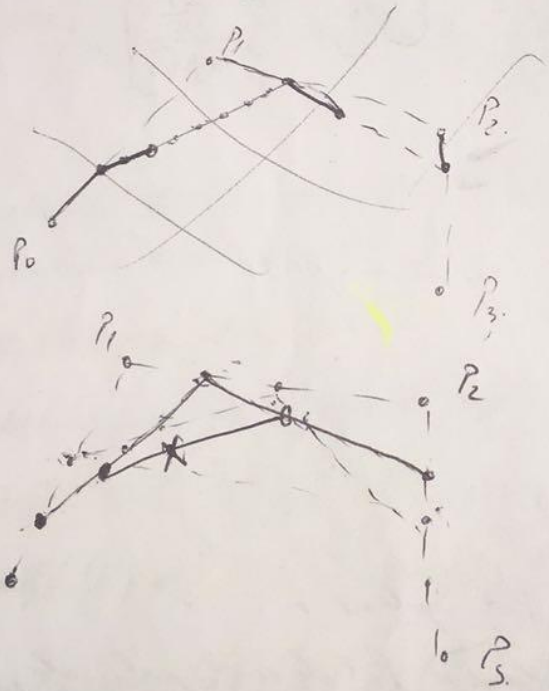
$\Rightarrow \langle f - g_1^*, g \rangle = \langle f - g_2^*, g \rangle$, $\forall g \in G$

~~$\langle f, g \rangle - \langle g_1^*, g \rangle$~~ Fie $g = g_1^* - g_2^*$

$\|g_1^* - g_2^*\| = \langle g_1^* - f + f - g_2^*, g \rangle = \langle f - g_2^*, g \rangle - \langle f - g_1^*, g \rangle = 0$

$\Rightarrow g_1^* = g_2^*$ → Unicitate

3/ Curbe Bézier.





$$\begin{cases} B_{11} = (1-t)\cdot P_0 + t \cdot P_1 \\ B_{12} = (1-t) P_1 + t \cdot P_2 \\ B_{23} = (1-t)\cdot P_2 + t \cdot P_3 \end{cases}$$

$\Rightarrow B_{02} = (1-t)\cdot B_{11} + B_{12}$

$B_{02} = (1-t)^2 \cdot P_0 + 2(1-t)tP_1 + t^2 P_2$

$\Rightarrow B_{13} = (1-t) B_{12} + t \cdot B_{23}$

$B_{13} = (1-t)^2 P_1 + 2(1-t)tP_2 + t^2 P_3$

$\Rightarrow B_{03} = (1-t) B_{02} + B_{13}$

Se calculează B_{02} și B_{13}

Derivare cu ajutorul Taylor, derivare cu pct de mijloc.

a) $f(x_0+h) = f(x_0) + f'(x_0)\cdot h + \frac{1}{2}\cdot f''(x_0)h^2 + \frac{1}{6}\cdot f'''(x_0)\cdot h^3 + \frac{1}{24}\cdot f^{(4)}(\xi)\cdot h^4 + \dots$

$f(x_0-h) = f(x_0) - f'(x_0)\cdot h + \frac{1}{2}\cdot f''(x_0)h^2 - \frac{1}{6}\cdot f'''(x_0)\cdot h^3 + \frac{1}{24}\cdot f^{(4)}(a)\cdot h^4$

$x_0-h < a < x_0 < b < x_0+h$

$f(x_0+h) + f(x_0-h) = 2f(x_0) + f''(x_0)\cdot h^2 + \left(\frac{1}{24}\cdot f^{(4)}(a)\cdot h^4 + \frac{1}{24}\cdot f^{(4)}(b)\cdot h^4\right)$

$f''(x_0) = \frac{1}{h^2}\underbrace{\left[f(x_0-h)+f(x_0+h)-2f(x_0)\right]}_{\text{cheia}} - \frac{h^2}{24}\cdot\left[f^{(4)}(a)+f^{(4)}(b)\right]$

$\Rightarrow \exists\, c\in(a,b)$ a.î. $f^{(4)}(c) = \frac{1}{2}\left[f^{(4)}(a)+f^{(4)}(b)\right]$

$\Rightarrow f''(x_0) = \frac{1}{h^2}\{\text{cheia}\} - \frac{h^2}{12}f^{(4)}(c)$

2016

1) Îmbunătățiți $N_1(h)$ care aprox pe M a.î să obț $N_2(h)$ care aprox cu o eroare $O(h^2)$

$M = N_1(h) + k_1 h + k_2 h^2 + \dots$ val pt $\forall h>0$

~~$M = N_1(\frac{h}{2}) + [N_1(\frac{h}{2}) - N_1(h)]$~~ e val și pt $\frac{h}{2}$

$M = N_1(\frac{h}{2}) + k_1\cdot\frac{h}{2} + k_2\cdot\frac{h^2}{4} + \dots$

$M = N_1(\frac{h}{2}) + \left[N_1(\frac{h}{2}) - N_1(h)\right] + k_2\cdot\left(\frac{h^2}{2}-h^2\right) + k_3\cdot\left(\frac{h^3}{4}-h^3\right) + \dots$

$\Rightarrow N_2(h) = N_1(\frac{h}{2}) + \left[N_1(\frac{h}{2}) - N_1(h)\right]$

$\Rightarrow M = N_2(h) - \underbrace{\frac{k_2}{2}h^2 - \frac{3}{4}k_3\cdot h^3 - \dots}_{\text{formată în } O(h^2)}$

2) $y' = y - t^2 + 1$, $t \in [0,2]$ — $f(y,t)$

$y(0) = 0.5$

$h = 0.5$

$y(1.3)$

$y(0.5) = y(0) + f(0, y(0)) \cdot h$.

$y(0.3) = 0.5 + 1.5 \cdot 0.5 = 1,25$

$y(1) = y(0.5) + f(0.5, y(0.5)) \cdot h$.

$y(1) = 1,25 + (1,25 - 0,25 + 1) \cdot 0,5 = 2,25$.

$y(1.5) = y(1) + f(1, y(1)) \cdot h = 3,375$.

$y = a_0 + a_1 x$.

$y(1) = a_0 + a_1 = 2,25$

$y(1.5) = a_0 + 1,5 a_1 = 3,375$

$0,5 a_1 = 1,125 \Rightarrow a_1 = 2,25$, $a_0 = 0$ $\Rightarrow y = 2,25 \cdot x$

$y(1,3) = 2,25 \cdot 1,3$

3) Explicați conceptul de cuadratură adaptivă.

4) Scrieți o fct care implementează trapez compus.

```
function rez = trapez(a, b, N, f)
    rez = f(a) + f(b)
    h = (b - a)/N
    for(i = 1 : N-1)
        rez += 2*f(a + i*h);
    endfor
    rez = rez * h/2;
endfunction
```

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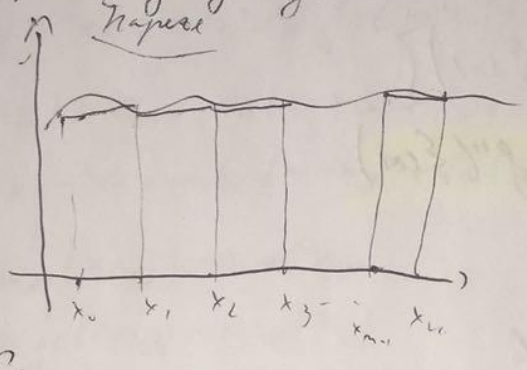
Se derivează

Examen

2016 → cealaltă var.

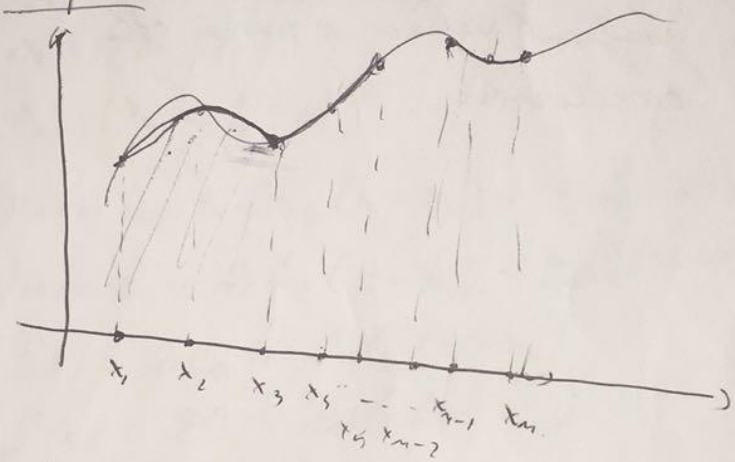
1) Explicați grafic formula compusă a trapezelor și Simpson

Trapeze



pct sunt echidistanți

Simpson



tot pct echidistanți

4) Deduceți folosind Lagrange Two point formula

$x_1, x_0, x_1 = x_0 + h$.

$$f(x) = L_1(x) + \frac{(x-x_0)(x-x_1)}{2!} f''(\xi(x))$$

$$L_1(x) = f(x_0) \cdot l_0(x) + f(x_1) \cdot l_1(x)$$

$$l_0(x) = \frac{x-x_1}{x_0-x_1}; \quad l_1(x) = \frac{x-x_0}{x_1-x_0} = \frac{x-x_0}{x_0+h-x_0} = \frac{x-x_0}{h}$$

$$l_0(x) = \frac{x-x_0-h}{x_0-x_0-h} = \frac{x-x_0-h}{-h}$$

$$f(x) = f(x_0) \cdot \frac{x-x_0-h}{-h} + f(x_1) \cdot \frac{x-x_0}{h} + \frac{1}{2}(x-x_0)(x-x_1-h) \cdot f''(\xi(x))$$

Se derivează

Examen

SD

$$f'(x) = \frac{f(x_0+h)-f(x_0)}{h} + \frac{1}{2}\{(x-x_0)\cdot 2-h\}\cdot f''(\xi(x)) +$$

$$+\frac{1}{2}(x-x_0)(x-x_0-h)\cdot[f''(\xi(x)]$$

$$f'(x_0) = \frac{f(x_0+h)-f(x_0)}{h} + \frac{h}{2} f''(\xi(x))$$

Idei bonus:

interpolări, statistici, cele vândute

integrare și derivare: grafică video și orice în afară accelerație

Examen

[redacted]

$f(x)$? $x = [0, 1, 2]$; $f = [6, 4, 3]$; $f' = [1, x, -1]$

$$\begin{cases} P_0(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 \\ P_1(x) = a_1 + b_1(x - x_1) + c_0(x - x_1)^2 + d_0(x - x_1)^3 \end{cases}$$

$$\begin{cases} P_{i+1}(x_{i+1}) = P_i(x_{i+1}) \\ P'_{i+1}(x_{i+1}) = P'_i(x_{i+1}) \\ P''_{i+1}(x_{i+1}) = P''_i(x_{i+1}) \\ P_i(x_i) = f(x_i) \\ P_{n-1}(x_n) = f(x_n) \end{cases}$$

$$P_0'(x) = b_0 + 2c_0(x - x_0) + 3d_0(x - x_0)^2$$

$$P_1'(x) = b_1 + 2c_1(x - x_1) + 3d_1(x - x_0)^2$$

$$P_0''(x) = 2c_0 + 6d_0(x - x_0)$$

$$P_1''(x) = 2c_1 + 6d_1(x - x_0)$$

$$P_0(x_0) = f(x_0) \Rightarrow a_0 = 6$$

$$P_1(x_1) = f(x_1) \Rightarrow a_1 = 4$$

$$P_1(x_2) = f(x_2) \Rightarrow a_1 + b_1 + c_1 + d_1 = 3$$

$$P_0(x_1) = P_1(x_1)$$

$$a_0 + b_0 + c_0 + d_0 = a_1 \Rightarrow b_0 + c_0 + d_0 = -2$$

$$P_0'(x_1) = P_1'(x_1)$$

$$b_0 + 2c_0 + 3d_0 = b_1 \Rightarrow b_0 - b_1 + 2c_0 + 3d_0 = 0$$

$$P_0''(x_1) = P_1''(x_1)$$

$$2c_0 + 6d_0 = 2c_1 \Rightarrow c_0 - c_1 + 3d_1 = 0$$

$$\begin{cases} b_1 + c_1 + d_1 = -1 \\ c_0 + d_0 = -3 \Rightarrow c_0 = -d_0 - 3 \Rightarrow d_0 = -c_0 - 3 \\ -b_1 + 2c_0 + 3d_0 = 1 \\ c_0 - c_1 + 3d_0 = 0 \\ b_1 + 2c_1 + 3d_1 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} b_1 + c_1 + d_1 = -1 \\ -b_1 + 2c_0 + 3(-c_0 - 3) = 1 \\ c_0 - c_1 + 3(-c_0 - 3) = 0 \\ b_1 + 2c_1 + 3d_1 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} b_1 + c_1 + d_1 = -1 \\ b_1 + 2c_1 + 3d_1 = 0 \\ -2c_0 - c_1 - 9 = 0 \Rightarrow -c_1 = 9 + 2c_0 \Rightarrow c_1 = -9 - 2c_0 \\ -b_1 - c_0 - 9 = 1 \Rightarrow b_1 = -c_0 - 10 \end{cases}$$

$$\Rightarrow \begin{cases} b_1 + c_1 + d_1 = -1 \\ c_1 + 2d_1 = 1 \\ b_1 = -(c_0 + 10) \\ c_1 = -(9 + 2c_0) \end{cases} \Rightarrow \begin{array}{l} -(c_0 + 10) - (9 + 2c_0) + d_1 = -1 \\ -(9 + 2c_0) + 2d_1 = 1 \end{array}$$

$$-3c_0 + d_1 = 18$$

$$2c_0 + 2d_1 = 10 \Rightarrow c_0 + d_1 = 5$$

$$c_0 = 5 - d_1$$

$$-3(5 - d_1) + d_1 = 18$$

$$-15 + 3d_1 + d_1 = 18$$

$$4d_1 - 15 = 18 \Rightarrow 4d_1 = 33$$

$$d_1 = \frac{33}{4}$$

1) Householder

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

$$E_1 = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix}.$$

$$\det(E_1) = (3-\lambda)^2 - 1 = 0$$

$$9 - 6\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - 6\lambda + 8 = 0 \quad \Delta = 36 - 32 = 4$$

$$\Delta = 4$$

$$\lambda_{1,2} = \frac{6 \pm \sqrt{4}}{2} \Rightarrow \lambda_1 = \frac{6+2}{2} = \frac{8}{2} = 4$$

~~$4 - 12 + 8$~~

$$\lambda_2 = \frac{6-2}{2} = \frac{4}{2} = 2.$$

$$\sqrt{=} 2$$

$$B = A - \sqrt{I_3} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$G_{21} = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad s = \sqrt{2}.$$

$$c = \frac{1}{\sqrt{2}} \; ; \; s = -\frac{1}{\sqrt{2}}$$

$$G_{21} \cdot B = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{array}\right] = \begin{bmatrix} \frac{2}{\sqrt{2}} & \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \end{bmatrix}.$$

$$G_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} \qquad s = 1.$$

$$c = 0 \; ; \; s = -1.$$

$$G_{32} \cdot G_{21} \cdot B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \cdot \left[\begin{array}{cc|c} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \end{array}\right] = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1. \\ 0 & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix}}_{R^{(1)}}.$$

$$Q^{(1)} = G_{32} \cdot G_{21}.$$

$$A^{(2)} = R^{(1)} \cdot Q^{(1)} + \sqrt{\lambda_1} I_n$$

$$\int_{-1}^{1} f(x)dx = a_1 f(x_1) + a_2 f(x_2) + a_3 f(x_3) + b_1 f(-1) + b_2 f(1)$$

Det x_i, a_i

b_j

a.i form. să aibe grad max de val.

$$\int_{-1}^{1} \pi(x) \cdot f(x) \cdot w(x) \cdot x^k dy$$

$f(x) = (x+1)(x-1)$.

$$\int_{-1}^{1} 1 dx = a_1 + a_2 + a_3 + b_1 + b_2 = 2$$

$$\int_{-1}^{1} x dx = a_1 x_1 + a_2 x_2 + a_3 x_3 + \cancel{b_1 x_1 + b_2 x_n}.$$

$$\int_a^b f(x)dx = \frac{b-a}{2} \int_{-1}^{1} f\left(\frac{b-a}{2}x + \frac{b-a}{2}\right)dx$$

$$I_{2n} + \frac{I_{2n} - I_n}{3} = I_{next}.$$

I_1 } . . .
I_2 } . . .
I_4 } . . .
I_8 } .

Spline-uri de clasă C_1

1. Rel. Hermite (2n+2)

$S_i(x_i) = f(x_i)$

$S_{n-1}(x_n) = f(x_n)$

$S_i'(x_i) = f'(x_i)$

$S_{n-2}'(x_n) = f'(x_n)$

2. Rel. Racordare.

$S_i(x_{i+1}) = S_{i+1}(x_{i+1})$

$S_i'(x_{i+1}) = S_{i+1}'(x_{i-1})$.

1) Householder

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

$$E_1 = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix}.$$

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$$c = \frac{1}{\sqrt{2}} \; ; \; s = -\frac{1}{\sqrt{2}}$$

$$G_{21} \cdot B = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{array}\right] = \begin{bmatrix} \frac{2}{\sqrt{2}} & \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \end{bmatrix}.$$

$$G_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix} \qquad s = 1.$$

$$c = 0 \; ; \; s = -1.$$

$$G_{32} \cdot G_{21} \cdot B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \cdot \left[\begin{array}{cc|c} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1 \end{array}\right] = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 1 & 1. \\ 0 & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix}}_{R^{(1)}}.$$

$$Q^{(1)} = G_{32} \cdot G_{21}.$$

$$A^{(2)} = R^{(1)} \cdot Q^{(1)} + \sqrt{\lambda_1} I_n$$

A = mată $m \times n$

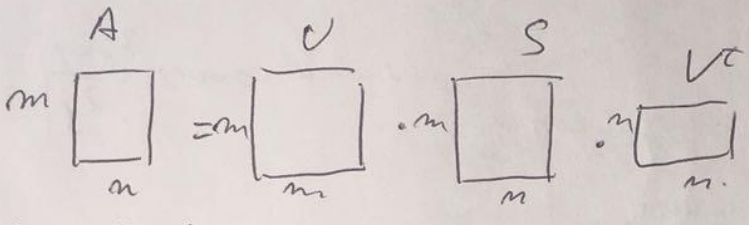
$A^T \cdot A$ și $A \cdot A^T$ simetrice

$Null(A) = Null(A^T \cdot A)$

$Rank(A) = Rank(A^T \cdot A)$

$sig(A^T \cdot A)$ și $sig(A \cdot A^T) \in \mathbb{R}^+$

Val proprii nenule ale sunt aceleași



Constr lui S

ρ_i^2 -> val proprii ale lui $A^T \cdot A$ => $S = \begin{bmatrix} s_1 & & & 0 \\ & s_2 & & \\ & & \ddots & \\ 0 & & & s_n \\ 0 & & & 0 \end{bmatrix}$

A = simetrică => $A = A^T$ => $A^2 = A^T A$

=> val singulare = val proprii

$\sqrt{\rho_i^2}$ -> val pozitive = val. singulare ale lui A

Constr lui V

M - simetrică => $M = V \cdot D \cdot V^t$, D = diag cu cont val proprii ale lui M, iar V = ortog cu col. vect proprii ale lui M

$A^T A$ = sim => $A^T \cdot A = V \cdot D \cdot V^T$

pt - că val proprii ale lui $A^T \cdot A > 0$ => $D = S^2$ -> el de pe diag la pătrat.

Constr. lui U

Fie $\rho_1 \geq \rho_2 \geq \dots \rho_k > 0$ și col lui V $v_1, v_2, \dots v_n$

$u_i = \frac{1}{\rho_i} \cdot A \cdot v_i$ => primele k col din U $[m \times m]$, mai lipsesc $m-k$ vect

îi alegem a.î matr să fie ortogonale și col lui indep cu Gram Schmidt

CURS-10 - interpolări

Weierstrass.

$|f(x) - P(x)| < \varepsilon$

Lagrange

$L_0(x) = \frac{x - x_1}{x_0 - x_1}$; $L_1(x) = \frac{x - x_0}{x_1 - x_0}$ — 2 puncte

? $L_0(x_0) = 0$; $L_1(x_1) = 1$.

$P(x) = L_0(x) f(x_0) + L_1(x) \cdot f(x_1)$

---

pt $n+1$ pct => polin. de grad n

$L_{n,k}(x_i) = 0$, $i \neq k$

$L_{n,k}(x_k) = 1$.

$$L_{n,k}(x) = \frac{(x - x_0) \cdots (x - x_{k-1}) \cdot (x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0) \cdots (x_k - x_{k-1}) \cdot (x_k - x_{k+1}) \cdots (x_k - x_n)}$$

$$P_n(x) = f(x_0) \cdot L_{n,0}(x) + \dots + f(x_n) \cdot L_{n,n}(x) = \sum_{k=0}^{n} f(x_k) \cdot L_{n,k}(x)$$

Eroarea $= \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \cdot (x - x_0)^{n+1}$

$$f(x) = P_n(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \cdot (x - x_0)(x - x_1) \cdots (x - x_n)$$

Lagrange Recursiv (Neville)

$$P(x) = \frac{(x - x_j) \cdot P_{0,1,\dots,j-1,j+1,\dots,n}(x) - (x - x_i) \cdot P_{0,1,\dots,i-1,i+1,\dots,n}(x)}{(x_i - x_j)}$$

$P_0 \to P_{0,1} \to P_{0,1,2} \to P_{0,1,2,3}$

$P_1 \to P_{1,2} \to P_{1,2,3}$

$P_2 \to P_{2,3}$

P_3

$P_{\emptyset} = \frac{x - x_i}{x_{i+1} - x_i}$

$$P_{01} = \frac{(x - x_1) \cdot P_0 - (x - x_0) \cdot P_1}{x_0 - x_1}$$

$P_{012} = P_{01} \cdot (x - x_2)$

Def divizate

cu def dele. => polin Newton

$$P_n(x) = f[x_0] + f[x_0,x_1](x-x_0) + f[x_0,x_1,x_2]\cdot(x-x_0)(x-x_1) + \dots + f[x_0,x_1,\dots,x_n]\cdot(x-x_0)\cdot(x-x_1)\cdot\dots\cdot(x-x_{n-1})$$

$$\Rightarrow P(x) = f[x_0] + \sum_{k=1}^{n} f[x_0,\dots,x_k]\cdot(x-x_0)\cdot\dots\cdot(x-x_{k-1}) \quad (*)$$

Dacă $h = x_{i+1} - x_i \Rightarrow x = x_0 + s\cdot h$

$$x - x_i = (s-i)\cdot h$$

$$\binom{s}{k} = \frac{s(s-1)\cdot\dots\cdot(s-k+1)}{k!} \rightarrow s \text{ nu e neapărat întreg}$$

$$P_n(x) = f[x_0] + \sum_{k=1}^{n} s(s-1)\cdot\dots\cdot(s-k+1)\cdot h^k\cdot f[x_0,x_1,\dots,x_k] \Big| \Rightarrow$$

$$\rightarrow P_n(x) = P_n(x_0+s\cdot h) = f[x_0] + \sum_{k=1}^{n}\binom{s}{k}\cdot k!\cdot h^k\cdot f[x_0,x_1,\dots,x_k]$$

Se mai $f[x_0,\dots,x_k] = \frac{1}{k!\cdot h^k}\cdot\Delta^k f(x_0)$

$$\Rightarrow P_n(x) = f[x_0] + \sum_{k=1}^{n}\binom{s}{k}\cdot\Delta^k f(x_0)$$

$$f[x_n,x_{n-1},\dots,x_{n-k}] = \frac{1}{k!h^k}\nabla^k f(x_n)$$

$$\binom{-s}{k} = (-1)^k\cdot\frac{s(s+1)\cdot\dots\cdot(s+k-1)}{k!}$$

$$P_n(x) = f[x_n] + \sum_{k=1}^{n}(-1)^k\cdot\binom{-s}{k}\nabla^k f(x_n)$$

(*)

$f[x_i] = f(x_i)$

$f[x_i,x_{i+1}] = \frac{f[x_{i+1}]-f[x_i]}{x_{i+1}-x_i}$

$f[x_i,\dots,x_{i+k}] = \frac{f[x_{i+1},\dots,x_{i+k}] - f[x_i,\dots,x_{i+k-1}]}{x_{i+k}-x_i}$

Hermite

~~Val lui f+~~

Dacă derivă polin Hermite = derivă lui f => $2n+1$ ec

Lagrange | => polin Hermite de gr. $2n+1$

$$H_{2n+1}(x) = \sum_{j=0}^{n} f(x_j)\cdot h_{n,j}(x) + \sum_{j=0}^{n} f'(x_j)\cdot\hat{H}_{n,j}(x)$$

$$h_{n,j} = [1 - 2(x-x_j)L'_{n,j}(x_j)]L^2_{n,j}(x)$$

$$\hat{H}_{n,j}(x) = (x-x_j)L^2_{n,j}(x)$$

~~$f(x)$~~ = Dacă $f \in C^{2n+2}[a,b]$

$$f(x) = H_{2n+1}(x) + \frac{f^{(2n+2)}(\xi(x))}{(2n+2)!}\cdot(x-x_0)^2\cdot\ldots\cdot(x-x_n)^2$$

$$f[x_0, x_1, \ldots, x_n] = \frac{f^{(n)}(\xi)}{n!}, \quad f \in C^n$$

Fie $z_{2k} = z_{2k+1} = x_k$

$$f[z_{2k}, z_{2k+1}] = \frac{f[z_{2k+1}] - f[z_{2k}]}{z_{2k+1} - z_{2k}}$$

$$f[x_k, x_{k+1}] = f'(\xi_k)$$

$$\lim_{x_{k+1}\to x_k} f[x_k, x_{k+1}] = f'(x_k) \quad \Big| \Rightarrow f[z_{2k}, z_{2k+1}] = f'(x_k)$$

Deci $H_{2n+1}(x) = f[z_0] + \sum_{k=1}^{2n+1} f[z_0, z_1, \ldots, z_k](x-z_1)\cdot\ldots\cdot(x-z_{k-1})$

Spline-uri

Liniar: Dacă știm f și f' → cond Hermite de continuitate la capete

Cubice

naturale	tensionat
$S''(x_0) = S''(x_n) = 0$	$S'(x_0) = f'(x_0)$
	$S'(x_n) = f'(x_n)$ dacă
	$S'(x_0) = f'(x_0)$

$$\begin{cases} S(x_j) = f(x_j), & j = 0, \ldots, n \\ S_{j+1}(x_{j+1}) = S_j(x_{j+1}), & j = 0, \ldots, n-2 \\ S'_{j+1}(x_{j+1}) = S'_j(x_{j+1}), & j = 0, \ldots, n-2 \\ S''_{j+1}(x_{j+1}) = S''_j(x_{j+1}), & j = 0, \ldots, n-2 \end{cases}$$

$$S_j(x) = a_j + b_j(x-x_j) + c_j(x-x_j)^2 + d_j(x-x_j)^3$$

Not $h_j = x_{j+1} - x_j$

$$a_{j+1} = S_{j+1}(x_{j+1}) = S_j(x_{j+1}) \Rightarrow \begin{cases} a_{j+1} = a_j + b_j h_j + c_j h_j^2 + d_j h_j^3 \\ b_{j+1} = b_j + 2c_j h_j + 3d_j h_j^2 \text{ (de înlocuit)} \end{cases}$$

din derivă 2 => $c_{j+1} = c_j + 3d_j h_j$ => d_j

$$\text{At: } \begin{cases} a_{j+1} = a_j + b_j h_j + \frac{h_j^2}{3}(2c_j + c_{j+1}) \\ b_{j+1} = b_j + h_j(c_j + c_{j+1}) \end{cases} \Rightarrow b_j$$

$$b_j = \frac{1}{h_j}(a_{j+1} - a_j) - \frac{h_j}{3}(2c_j + c_{j+1})$$

$$\Rightarrow h_{j-1} \cdot c_{j-1} + 2(h_{j-1} + h_j) \cdot c_j + h_j \cdot c_{j+1} = \underbrace{\frac{3}{h_j}(a_{j+1} - a_j) - \frac{3}{h_{j-1}}(a_j - a_{j-1})}_{\text{col}}$$

=> sistem tridiagonal. se află c_j și se scot restul care depind doar de c_j

Curbe ~~Bézier Hermite~~

Polin Bernstein $B_i^m(t) = \binom{m}{i} t^i \cdot (1-t)^{m-i}$

Prop: $B_i^m > 0$

$$\sum_{i=0}^{m} B_i^m(t) = 1.$$

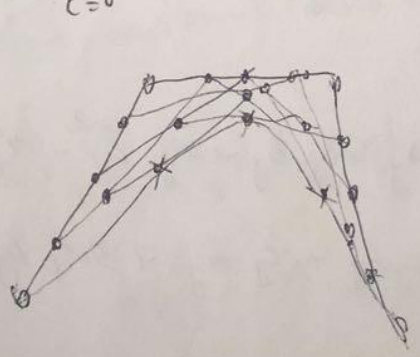
Simetrie: $B_i^m(t) = B_{m-i}^m(1-t)$

Max unic: $t = \frac{i}{m}$ -> $t \in [0,1]$.

Rec: $B_i^m(t) = (1-t) \cdot B_i^{m-1}(t) + t \cdot B_{i-1}^{m-1}(t)$

Curbe Bézier

$$B(t) = \sum_{i=0}^{m} P_i \cdot B_i^m(t)$$





Forma matr. Lagr.

$$V = \begin{bmatrix} 1 & a-x_1 & \cdots & a-x_1 \\ \vdots & & & \\ a-x_m & a-x_m & \cdots & 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & x_1-x_1 & \cdots & x_m-x_1 \\ \vdots & & & \\ x_1-x_m & x_2-x_m & \cdots & 1 \end{bmatrix}$$

$L = \text{prod}(V).\ /\ \text{prod}(U)$

$b = \sum_{i=1}^{m} l_i \cdot y_i = \text{prod}(V).\ /\ \text{prod}(U) * y.$

$U = \text{diag}(x) * \text{ones}(m) - \text{ones}(m) * \text{diag}(x) + \text{eye}(m)$

$V = (a - \text{diag}(x)) * \sim \text{eye}(m) + \text{eye}(m)$

Spline liniar.

$P_i(x) = a_i x + b_i$

cond de interpolare:

$P_i(x_i) = f(x_i), \ i = 0 : m-1$

$P_{m-1}(x_m) = f(x_m)$

cond de racordare:

$P_i(x_{i+1}) = P_{i+1}(x_{i+1})$

$$\Rightarrow a_i = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

$$b_i = \frac{x_{i+1} \cdot f(x_i) - x_i \cdot f(x_{i+1})}{x_{i+1} - x_i}$$

La Spline-uri (clasă C1) se folosesc bază Bernstein pt simplif vol de calcul,

bază Bernstein: $(1-t)^3,\ 3t(1-t)^2,\ 3t^2(1-t),\ t^3$

$$P_i(t) = a'_i(1-t)^3 + 3b'_i(1-t)^2 t + 3c'_i t^2(1-t) + d'_i t^3$$

$a'_i = f(x_i)$

$b'_i = f(x_i) + \frac{h_i}{3} \cdot f'(x_i)$ $\quad h_i = x_{i+1} - x_i$

$c'_i = f(x_{i+1}) - \frac{h_i}{3} f'(x_{i+1})$

$d'_i = f(x_{i+1})$

$\Delta = 4$

$\Rightarrow z_1 = \frac{6+2}{2} = \frac{8}{2} = 4$

Spline natural C^2

$$A = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ h_0 & 2(h_0+h_1) & h_1 & \\ 0 & h_1 & 2(h_1+h_2) & h_2 \\ & & \ddots & \ddots \\ & & h_{n-2} & 2(h_{n-2}+h_{n-1}) \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad X = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix}$$

Spline natural C_2.

$$A = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ h_0 & 2(h_0+h_1) & h_1 & & \\ 0 & h_1 & 2(h_1+h_2) & h_2 & \\ & & \ddots & \ddots & \ddots \\ & & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ \frac{3}{h_{i+1}}(a_{i+1}-a_i) - \frac{3}{h_i}(a_i-a_{i-1}) \\ \frac{3}{h_1}(a_2-a_1) - \frac{3}{h_0}(a_1-a_0) \\ \vdots \\ 0 \end{bmatrix}$$

$$\frac{3}{h_1}(a_2-a_1) - \frac{3}{h_0}(a_1-a_0)$$

Tensional

$$A = \begin{bmatrix} 2h_0 & h_0 & 0 & \cdots \\ & \text{La fel.} & & \\ 0 & h_{n-1} & 2h_{n-1} & \end{bmatrix}; \quad b = \begin{bmatrix} \frac{3}{h_0}(a_1-a_0) - 3f'(a) \\ \text{La fel.} \\ 3f'(b) - \frac{3}{h_{n-1}}(a_n-a_{n-1}) \end{bmatrix}$$

8 6

CURS 10

Eroarea dreptei care trece printre pct

$$E_1(a_0,a_1) = \sum_{i=1}^{10} |y_i - (a_1 x_i + a_0)|$$

$$E_1 = \min \Leftrightarrow \begin{cases} \frac{\partial}{\partial a_0} E_1(a_0,a_1) = 0 \\ \frac{\partial}{\partial a_1} E_1(a_0,a_1) = 0 \end{cases} \rightarrow \text{fct nu e derivabila in 0.}$$

$$\Rightarrow E_2(a_0,a_1) = \sum_{i=1}^{10} |y_i - (a_1 x_i + a_0)|^2 = \sum_{i=1}^{10} (y_i - (a_1 x_i + a_0))^2$$

$$\begin{cases} \frac{\partial E_2(a_0,a_1)}{\partial a_0} = 2 \sum_{i=1}^{m} (y_i - a_1 x_i - a_0) \cdot (-1) = 0 \\ \frac{\partial E_2}{\partial a_1} = 2 \sum_{i=1}^{m} (y_i - a_1 x_i - a_0) \cdot (-x_i) \end{cases}$$

$$\Rightarrow \begin{cases} a_0 \cdot \sum_{i=1}^{m} x_i + a_1 \sum_{i=1}^{m} x_i^2 = \sum_{i=1}^{m} x_i y_i \\ a_0 \cdot m + a_1 \cdot \sum_{i=1}^{m} x_i = \sum_{i=1}^{m} y_i \end{cases} \Rightarrow a_0, a_1$$

$P_n(x) = a_n \cdot x^n + a_{n-1} \cdot x^{n-1} + \dots + a_1 x + a_0$; aprox a m pct cu un polin de grad n, $n < m-1$.

$$E_2 = \sum_{i=1}^{m} (y_i - P_n(x_i))^2$$

$\frac{\partial E_2}{\partial a_j} = 0$ pt a minimiza pe n.

$$\begin{cases} a_0 \cdot \sum_{i=1}^{m} x_i^0 + a_1 \sum_{i=1}^{m} x_i^1 + \dots + a_n \cdot \sum_{i=1}^{m} x_i^n = \sum_{i=1}^{m} y_i x_i^0 \\ \vdots \\ a_0 \cdot \sum_{i=1}^{m} x_i^n + a_1 \sum_{i=1}^{m} x_i^{n+1} + \dots + a_n \cdot \sum_{i=1}^{m} x_i^{2n} = \sum_{i=1}^{m} y_i x_i^n \end{cases}$$

S.L.E.

Aprox continuă în sensul cmmp.

$P_m(x) = a_m x^m + a_{m-1} x^{m-1} + \dots + a_1 x + a_0 = \sum_{k=0}^{m} a_k \cdot x^k$

Pt a min err.

$E(a_0, \dots, a_m) = \int_a^b (f(x) - P_m(x))^2 dx = \int_a^b \left(f(x) - \sum_{k=0}^{m} a_k \cdot x^k \right)^2 dx$

$\frac{\partial E}{\partial a_j}(a_0, \dots, a_m) = 0 \rightarrow$ cond necesară

$E = \int_a^b (f(x))^2 dx - 2 \sum_{k=0}^{m} a_k \cdot \int_a^b x^k \cdot f(x) dx + \int_a^b \left(\sum_{k=0}^{m} a_k \cdot x^k \right)^2 dx$

(x desfac parantezele)

// toate a_i ți a_j sunt 0, $a_j = 1$

$\frac{\partial E}{\partial a_j}(a_0, \dots, a_m) = -2 \int_a^b x^j \cdot f(x) dx + 2 \sum_{k=0}^{m} a_k \int_a^b x^{j+k} dx = 0$

$\Rightarrow \sum_{k=0}^{m} a_k \cdot \int_a^b x^{j+k} dx = \int_a^b x^j f(x) dx \quad , j = 0 - n$

$\rightarrow$ Matr sist = Matr Hilbert = ~~problema de~~ rău condiționată

$\{\phi_0, \phi_1, \dots, \phi_n\}$ fct lin indep $\Leftrightarrow c_0 \phi_0(x) + c_1 \phi_1(x) + \dots + c_n \phi_n(x) = 0$

$\Leftrightarrow c_0 = c_1 = \dots = c_n = 0$

1) $\phi_j(x)$ = polin de grad $j \Rightarrow$ fct lin indep

(+ cuadratură = comb liniară de $\phi_j(x)$ $j = 0 - n$

(quadratic polynomial)

$w(x) = \frac{1}{\sqrt{1-x^2}} \rightarrow$ fct de greutate

$P(x) = \sum_{k=0}^{n} a_k \cdot \phi_k(x) \Rightarrow E(a_0, \dots, a_n) = \int_a^b w(x) \cdot (f(x) - P(x))^2 dx.$

$$0 = \frac{\partial E}{\partial a_j} = 2 \int_a^b w(x) \cdot \left(f(x) - \sum_{k=0}^{n} a_k \phi_k(x) \right) \cdot \phi_j(x) \, dx$$

$$\Leftrightarrow \int_a^b w(x) \cdot f(x) \cdot \phi_j(x) \, dx = \sum_{k=0}^{n} a_k \cdot \int_a^b w(x) \cdot \phi_k(x) \cdot \phi_j(x) \, dx$$

Știm să aliniem ϕ_j a.î.

$$\int_a^b w(x) \cdot \phi_k(x) \phi_j(x) \, dx = \begin{cases} 0, & j \neq k \\ \alpha_k > 0, & j = k \end{cases}$$

$$\int_a^b w(x) f(x) \cdot \phi_j(x) \, dx = a_j \int_a^b w(x) \cdot [\phi_j(x)]^2 \, dx = a_j \cdot \alpha_j$$

$$\Rightarrow a_j = \frac{1}{\alpha_j} \cdot \int_a^b w(x) \cdot f(x) \cdot \phi_j(x) \, dx$$

Dacă $\alpha_k = 1 \Leftrightarrow \{\phi_1, \dots, \phi_n\}$ ortonormate

$\Rightarrow$ Dacă $\{\phi_0, \dots, \phi_n\}$ ortonormate

$$P(x) = \sum_{k=0}^{n} a_k \cdot \phi_k(x), \text{ cu } a_k = \frac{\int_a^b w(x) \cdot \phi_k(x) \cdot f(x) \, dx}{\int_a^b w(x) \cdot (\phi_k(x))^2 \, dx}$$

$$\Rightarrow a_k = \frac{1}{\alpha_k} \int_a^b w(x) \cdot \phi_{(k)}(x) \cdot f(x) \, dx$$

$\phi_0(x) = 1$

$\phi_1(x) = x - B_1$, $\quad B_k = \dfrac{\int_a^b x \, w(x) \cdot (\phi_{k-1}(x))^2 \, dx}{\int_a^b w(x) \cdot (\phi_{k-1}(x))^2 \, dx}$

$k \geq 2$, $\phi_k(x) = (x - B_k) \cdot \phi_{k-1}(x) - C_k \cdot \phi_{k-2}(x)$

$$C_k = \frac{\int_a^b x \, w(x) \cdot \phi_{k-1}(x) \cdot \phi_{(k-2)}(x) \, dx}{\int_a^b w(x) \cdot (\phi_{k-2}(x))^2 \, dx}$$

Polin Legendre

$P_1(x) = x$

$P_2(x) = x^2 - \frac{1}{3}$ ...

$$P_{n+1}(x) = x \cdot P_n(x) - \frac{n^2}{4n^2-1} \cdot P_{n-1}(x)$$

Polin Chebyshev

$T_n(x)$ ortog. pe $(-1,1)$ cu fct $\frac{1}{\sqrt{1-x^2}}$ $\Rightarrow \int_{-1}^{1} \frac{T_i(x) \cdot T_j(x)}{\sqrt{1-x^2}} dx = \begin{cases} 0, & i \neq j \\ \frac{\pi}{2}, & i = j \neq 0 \\ \pi, & i = j = 0 \end{cases}$

$T_n(x) = \cos(n \arccos(x))$

$T_{n+1}(x) = 2x T_n(x) - T_{n-1}(x)$

x not $\arccos(x) = \theta$

by x obtained

Polin chebyshev de grad ≥ 1 au n 0-uri ci

$\bar{x}_k = \cos\left(\frac{2k-1}{2n}\pi\right)$ $\rightarrow$ x aleg pct echidistante pt a minimiza eroarile

Aprox cu fct rationale

$f(x) \approx r(x) = \frac{p(x)}{q(x)}$

Aprox Padé aleg $N+1$ parametri a.i. $f^{(k)}(0) = r^{(k)}(0)$

$$0 = f(x) - r(x) = f(x) - \frac{p(x)}{q(x)} = \frac{f(x) \cdot q(x) - p(x)}{q(x)} = \frac{f(x) \cdot \sum_{i=0}^{m} q_i x^i - \sum_{i=0}^{n} p_i x^i}{q(x)}$$

Daca $f(x)$ se poate scrie ca serie Maclaurin, $f(x) = \sum_{i=0}^{\infty} a_i x^i$

$$f(x) - r(x) = \frac{\sum_{i=0}^{\infty} a_i x^i \cdot \sum_{i=0}^{m} q_i x^i - \sum_{i=0}^{n} p_i x^i}{q(x)} \quad | : x^k$$

$\simeq 0$ (pt că $f(x) \approx r(x)$)

luăm $p_{n+1} = p_{n+2} = \dots = p_N = 0$

$q_{m+1} = q_{m+2} = \dots = q_N = 0$, $m + n = N + 1$ necunoscute

Şi se cere $\forall k$ $\sum_{i=0}^{k} a_i q_{k-i} = p_k$, $k = 0, 1, \dots, N$.

Aprox cea polin trigon.

$\{\phi_0, \phi_1, \dots, \phi_{2n-1}\}$; $\phi_0(x) = \frac{1}{2}$; $\phi_k(x) = \cos(kx)$, $k = 1, \dots, n$

↳ ortogonalepe $[-\pi, \pi]$.

$\phi_{n+k}(x) = \sin(kx)$, $k = 1, \dots, n-1$

$$\int_{-\pi}^{\pi} \phi_{n+k}(x) \cdot \phi_j(x)\,dx = \int_{-\pi}^{\pi} \sin(kx) \cdot \cos(jx)\,dx = 0$$

$\sin(kx)\cdot\cos(jx) = \frac{1}{2}\left(\sin((k+j)x) + \sin((k-j)x)\right) \neq 0$

$$S_n(x) = \frac{1}{2}a_0 + \sum_{k=1}^{2n-1} a_k \cdot \phi_k(x)$$

$$a_k = \frac{1}{\pi} \cdot \int_{-\pi}^{\pi} f(x) \cdot \phi_k(x)\,dx$$

$$E(S_n) = \sum_{j=0}^{2n-1} (y_j - S_n(x_j))^2$$

$$= \sum_{j=0}^{2n-1} \left(y_j - \left(\frac{a_0}{2} + a_n \cos(nx_j) + \sum_{k=1}^{n-1} \left(a_k \cos(kx_j) + b_k \sin(kx_j) \right) \right) \right)^2$$

$\sum_{j=0}^{2n-1} \phi_k(x_j) \cdot \phi_l(x_j) = 0 \Rightarrow E(a_0, \dots, a_n, b_1, \dots, b_{n-1}) = \sum_{j=0}^{2n-1} (y_j - S_n(x_j))^2$

$$a_k = \frac{1}{n} \cdot \sum_{j=0}^{2n-1} y_j \cdot \cos(kx_j)$$

$$b_k = \frac{1}{n} \cdot \sum_{j=0}^{2n-1} y_j \cdot \sin(kx_j)$$

FFT

$$S_n(x) = \frac{a_0}{2} + a_n \cos(nx) + \sum_{k=1}^{n-1} (a_k \cos kx + b_k \sin(kx))$$

~~$f(x) = \frac{1}{n} \cdot e^{iz} = \cos(z) + i\sin(z)$~~ (ρ^e pg 14)

$$\frac{1}{n} \cdot c_k \, e^{-i\pi k} = \frac{1}{n} \cdot \sum_{j=0}^{2n-1} y_j \cdot e^{\frac{\pi i j k}{n}} \cdot e^{-i\pi k} = \frac{1}{n} \cdot \sum_{j=0}^{2n-1} y_j \cdot e^{ik\left(-\pi + \frac{\pi j}{n}\right)}$$

$$= \frac{1}{n} \cdot \sum_{j=0}^{2n-1} \cdot y_j \left(\cos kx_j + i \sin kx_j \right) \Rightarrow \frac{1}{n} \cdot c_k \cdot e^{-i\pi k} = a_k + i b_k$$

$\Rightarrow \lambda_1 = \frac{6+2}{2} = \frac{8}{2} = 4$

Din curs

$\|f - g^*\| = \min_{g \in G} \|f - g\|$ – se-arr facut cu Rodin bere

Dem. $\langle f - g^*, g \rangle = 0$, g^* e unic.

$\langle f - g^*, g \rangle = \langle f, g \rangle - \langle g^*, g \rangle = 0$

$\Rightarrow \langle f, g \rangle = \langle g^*, g \rangle$

Presp. ca $\exists g_1^*$ si $g_2^* \Rightarrow \langle g_1^*, g \rangle = \langle g_2^*, g \rangle$

$\Rightarrow g_1^* = g_2^*$

La FFT

$$c_k + c_{m+k} = \sum_{j=0}^{2m-1} y_j \cdot \ell^{\frac{ik\bar{\pi}j}{m}} + \sum_{j=0}^{2m-1} y_j \cdot \ell^{\frac{i(m+k)\bar{\pi}j}{m}}$$

$$= \sum_{j=0}^{2m-1} y_j \cdot \ell^{\frac{ik\bar{\pi}j}{m}} \cdot (1 + \ell^{\bar{\pi}ij})$$

$$1 + \ell^{\bar{\pi}ij} = \begin{cases} 2, & j \text{ par} \\ 0, & j = \text{impar} \end{cases}$$

$c_1, c_2, \dots, c_m$

c_k, c_{m+k}

c_k

$\log(n)$

$c(m) \to c(\frac{m}{2})$

Trans $j \to 2j$

$$\Rightarrow \begin{cases} c_k + c_{m+k} = 2 \cdot \sum_{j=0}^{m-1} y_{2j} \cdot \ell^{\frac{ik\bar{\pi}2j}{m}} \\ c_k - c_{m+k} = 2 \cdot \ell^{\frac{ik\bar{\pi}}{m}} \cdot \sum_{j=0}^{m-1} y_{2j+1} \ell^{\frac{ik\bar{\pi}j \cdot 2}{m}} \end{cases}$$

$c_k \leftarrow c_{m+k}$ m = putere de 2.

$c_1 \to c_2 \to$

$1 \to (m+1)$

$$\begin{cases} c_k + c_{m+k} = \dots \\ c_k - c_{m+k} = \dots \end{cases}$$

$$\begin{cases} c_k + c_{\frac{m}{2}+k} = \dots \\ c_k - c_{\frac{m}{2}+k} = \dots \end{cases}$$

$$\begin{cases} c_k + c_{\frac{m}{4}+k} = \dots \\ c_k - c_{\frac{m}{4}+k} = \dots \end{cases}$$

CURS 11 $x_1 = x_0 + h$.

$$f(x) = \underbrace{\frac{f(x_0)\cdot(x-x_0-h)}{-h} + \frac{f(x_0+h)(x-x_0)}{h}}_{\text{Polin Lagrange}} + \underbrace{\frac{(x-x_0)(x-x_0-h)}{2}\cdot f''(\xi(x))}_{\text{Eroare}}$$

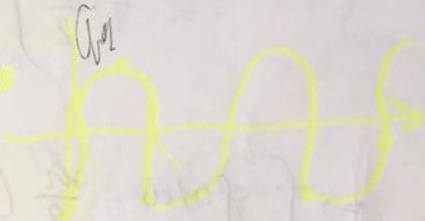
$$f'(x) = \frac{f(x_0+h)-f(x_0)}{h} + \frac{2(x-x_0)-h}{2} f''(\xi(x)) + \Delta_x\left[\frac{(x-x_0)(x-x_0-h)}{2} f''(\xi(x))\right]$$

$$\Rightarrow f'(x) \approx \frac{f(x_0+h)-f(x_0)}{h}, \text{ val}$$

$$f'(x) = \frac{f(x_0+h)-f(x_0)}{h} + \frac{2(x-x_0)-h}{2} f''(\xi(x)) + \frac{(x-x_0)(x-x_0-h)}{2}\cdot \Delta_x\cdot(f''(\xi(x)))$$

Eroare.

$$\Rightarrow f'(x_0) = \frac{f(x_0+h)-f(x_0)}{h} - \frac{h}{2} f''(\xi)$$



Fol. pol Lagrange

$$f(x) = \sum_{j=0}^{n} f(x_j)\cdot L_j(x) + \frac{(x-x_0)\cdot\ldots\cdot(x-x_n)}{(n+1)!} f^{(n+1)}(\xi(x))$$

$$f'(x) = \sum_{j=0}^{n} f(x_j)\cdot L_j'(x) + \frac{f^{(n+1)}(\xi(x_a))}{(n+1)!}\cdot \prod_{\substack{j=0 \\ j\neq k}}^{n} (x_a - x_j)$$

Formula in 3 pct.

$$f'(x_0) = \frac{1}{2h}\left[-3f(x_0) + 4f(x_0+h) - f(x_0+2h)\right] + \frac{h^2}{3} f'''(\xi)$$

(x_0, x_0+h, x_0+2h).

sau

$$f'(x_0) = \frac{1}{2h}\cdot\left[f(x_0+h) - f(x_0-h)\right] - \frac{h^2}{6}\cdot f'''(\xi)$$

(x_0-h, x_0, x_0+h).

Integrare numerică

$$\int_a^b f(x)\,dx = \int_a^b \sum_{i=0}^{n} f(x_i)\cdot L_i(x)\,dx + \int_a^b \prod_{i=0}^{n}(x-x_i)\,\frac{f^{(n+1)}\cdot\xi(x)}{(n+1)!}\,dx$$

$$a_i = \int_a^b L_i(x)\,dx \Rightarrow \int_a^b f(x)\,dx \simeq \sum a_i \cdot f(x_i)$$

$$\text{err}$$

$$E(f) = \frac{1}{(n+1)!}\cdot\int_a^b \prod (x-x_i)\cdot f^{(n+1)}(\xi(x))\,dx$$

Reg trapezelor.

$$\int_{x_0}^{x_1} f(x)\,dx = \int_{x_0}^{x_1}\left[\frac{(x-x_1)}{(x_0-x_1)}\cdot f(x_0) + \frac{(x-x_0)}{(x_1-x_0)} f(x_1)\right]dx$$

$$+ \frac{1}{2}\int_{x_0}^{x_1} f''(\xi(x))(x-x_0)(x-x_1)\,dx$$

$$\text{err} = -\frac{h^3}{6} f''(\xi)$$

$$\Rightarrow \int_a^b f(x)\,dx = \frac{h}{2}[f(x_0)+f(x_1)] - \frac{h^3}{12} f''(\xi)$$

$x_1 = x_0 + h$

Reg Simpson:

$$\int_a^b f(x)\,dx = \frac{b-a}{6}\cdot\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right] - \frac{f^{(4)}(\xi)}{2880}(b-a)^5$$

Dem

$$\int_a^b f(x)\,dx = \int_{x_0}^{x_2} \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) +$$

→ fol Lagrange

$$+ \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2)\Big]dx + \int_{x_0}^{x_2} \frac{(x-x_0)(x-x_1)(x-x_2)}{6} f^{(3)}(\xi(x))$$

Taylor:

$$f(x) = f(x_1) + f'(x_1)(x-x_1) + \frac{f''(x_1)}{2}(x-x_1)^2 + \frac{f'''(x_1)}{6}(x-x_1)^3 + \frac{f^{(4)}(\xi(x))}{24}(x-x_1)^4$$

Integrând pe Taylor.

$$\int_{x_0}^{x_2} f(x)\,dx = \left(f(x_1)(x-x_1) + \frac{f'(x_1)}{2}(x-x_1)^2 + \frac{f''(x_1)}{6}(x-x_1)^3 + \right.$$

$$\left. + \frac{f'''(x_1)}{24}(x-x_1)^4 \right\}\Big|_{x_0}^{x_2} + \frac{1}{24}\int_{x_0}^{x_2} f^{(4)}(\xi(x_1))(x-x_1)^4\,dx$$

$h = x_2 - x_1 = x_1 - x_0$

$$\int_{x_0}^{x_2} f(x)\,dx = 2hf(x_1) + \frac{h^3}{3}f''(x_1) + \frac{f^{(4)}(\xi_1)}{60}h^5$$

→ produce fără Taylor $f(x_0+h) + f(x_0-h)$

Dacă aprox pe $f''(x_1) = \frac{1}{h^2}\left(f(x_0) - 2f(x_1) + f(x_2) \right\} - \frac{h^2}{12}f^{(4)}(\xi_2)$

$$\int_{x_0}^{x_2} f(x)\,dx = 2hf(x_1) + \frac{h^3}{3}\left[\frac{1}{h^2}\left\{ f(x_0) - 2f(x_1) + f(x_2) \right] - \frac{h^2}{12}f^{(4)}(\xi_2) \right] +$$

$$+ \frac{f^{(4)}(\xi_1)}{60}h^5$$

$$= \frac{h}{3}\cdot\left[f(x_0) + 4f(x_1) + f(x_2) \right] - \frac{h^5}{90}\cdot f^{(4)}(\xi) \Rightarrow \text{Simpson}$$

Newton-Cotes

Închise $\to n+1$ pct $x_i = x_0 + ih$, $i = 0, \dots, n$, $x_0 = a$, $x_n = b$, $h = \frac{b-a}{n}$.

$$\int_a^b f(x)\,dx \simeq \sum_{i=0}^{n} a_i \cdot f(x_i)$$

$$a_i = \int_{x_0}^{x_n} L_i(x)\,dx = \int_{x_0}^{x_n} \prod_{\substack{j=0 \\ j\neq i}}^{n} \frac{(x-x_j)}{(x_i-x_j)}\,dx.$$

Deschise $\int_a^b f(x)\,dx = \int_{x_{-1}}^{x_{n+1}} f(x)\,dx \simeq \sum_{i=0}^{n} a_i \cdot f(x_i)$

$$a_i = \int_a^b L_i(x)$$

$x_i = x_0 + ih$.

$i = 0, \dots, n+1$.

$h = \frac{b-a}{n+2}$, $x_0 = a+h$, $x_n = b-a$.

$x_{-1} = a$, $x_{n+1} = b$.

Formula compusă → se împarte intervalul în n subintervale pe care se aplică Simpson/Trapez

Midpoint rule

$$\int_a^b f(x)\,dx = (b-a)\cdot f\left(\frac{a+b}{2}\right) + \frac{f''(\xi)}{24}(b-a)^3$$

Regula compusă

Trapez

$$\int_a^b f(x)\,dx = \frac{h}{2}\left[f(a) + 2\sum_{j=1}^{n-1} f(x_j) + f(b)\right] - \frac{b-a}{12}h^2 \cdot f''(\mu)$$

Trapez cu pct de mij

$$\int_a^b f(x)\,dx = 2h \cdot \sum_{j=0}^{\frac{n}{2}} f(x_{2j}) + \frac{b-a}{6}\cdot h^2 f''(\mu)$$



Simpson

Alegem n par => 2n pct.

$$\int_a^b f(x)\,dx = \sum_{j=1}^{\frac{n}{2}} \int_{x_{2j-2}}^{x_{2j}} f(x)\,dx = \sum_{j=1}^{\frac{n}{2}} \left\{\frac{h}{3}\left[f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j})\right] - \frac{h^5}{90} f^{(4)}(\xi_j)\right.$$

Ținând cont că se repetă termenii:

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[f(x_0) + 2\cdot\sum_{j=1}^{\frac{n}{2}-1} f(x_{2j}) + 4\sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) + f(x_n)\right] - \frac{h^5}{90}\sum_{j=1}^{\frac{n}{2}} f^{(4)}(\xi_j)$$

8.50 — 10

x — 3 => $x = \frac{3\cdot 8.50}{10} = \frac{25,5}{10} = 2,55$

Curs 12

~~Integrare~~ Romberg

~~Extra~~ Extrapolare Richardson

$M - N(h) = k_1 h + k_2 h^2 + k_3 h^3 + \dots$

$$\begin{cases} M = N(h) + k_1 h + k_2 h^2 + k_3 h^3 + \dots \\ M = N\left(\frac{h}{2}\right) + k_1 \cdot \frac{h}{2} + k_2 \cdot \frac{h^2}{4} + \dots \end{cases} \quad 1 \cdot 2 \ominus$$

$$M = \left[N\left(\frac{h}{2}\right) + \left(N\left(\frac{h}{2}\right) - N(h)\right)\right] + k_2\left(\frac{h^2}{2} - h^2\right) + k_3\left(\frac{h^3}{4} - h^3\right) + \dots$$

$$N_2(h) = N_1\left(\frac{h}{2}\right) + \left[N_1\left(\frac{h}{2}\right) - N_1(h)\right]$$

$$\Rightarrow M = N_2(h) - \frac{k_2}{2} \cdot h^2 - \frac{3k_3}{4} h^3 - \dots$$

La trapeze:

$$\int_a^b f(x)\,dx = \frac{h}{2}\left[f(a) + f(b) + 2\sum_{j=1}^{m-1} f(x_j)\right] - \frac{(b-a)}{12} h^2 \cdot f''(\mu)$$

$h = \frac{b-a}{m}$

$x_j = a + jh$

$R_{11}, R_{21}, R_{31}, \dots$ aprox integrale pt 1, 2, 4, 8 ... puncte

$$\Rightarrow R_{k,2} = R_{k,1} + \frac{1}{3}\left(R_{k,1} - R_{k-1,1}\right) \qquad k = 2, 3 \dots \quad O(h^4)$$

$$R_{k,3} = R_{k,2} + \frac{1}{15}\left(R_{k,2} - R_{k-1,2}\right) \qquad k = 3, 4 \dots \quad O(h^6)$$

$$R_{k,j} = R_{k,j-1} + \frac{1}{4^{j-1} - 1}\left(R_{k,j-1} - R_{k-1,j-1}\right) \qquad k = j, j+1, \dots$$

$$\begin{cases} R_{k,1} = \frac{1}{2}\left[R_{k-1,1} + h_{k-1} \sum_{i=1}^{2^{k-2}} f\left(a + (2i-1)h_k\right)\right] \\ R_{11} = R_1 = \frac{b-a}{2}\left[f(a) + f(b)\right] \\ R_{21} = \frac{1}{2}\left[R_{1,1} + h_1 f(a + h_2)\right] \\ R_{31} = \frac{1}{2}\left[R_{2,1} + h_2\left(f(a + h_3) + f(a + 3h_3)\right)\right] \end{cases}$$

Cuadraturi adaptive

Simpson

$$\int_a^b f(x)dx = S(a,b) - \frac{h^5}{90} f^{(4)}(\xi)$$

$$S(a,b) = \frac{h}{3}\cdot[f(a)+4f(a+h)+f(b)] \rightarrow \text{schimbăm pasul cu } \frac{h}{2}$$

$$\int_a^b f(x)dx = \frac{h}{6}[f(a)+4f(a+\frac{h}{2})+2f(a+h)+4f(a+\frac{3h}{2})+f(b)] - (\frac{h}{2})^4 \cdot (\frac{b-a}{180}) \cdot f^{(4)}(\xi)$$

Not cu

$$S(a, \frac{a+b}{2}) = \frac{h}{6}[f(a)+4f(a+\frac{h}{2})+f(a+h)]$$

$$S(\frac{a+b}{2}, b) = \frac{h}{6}[f(a+h)+4f(a+\frac{3h}{2})+f(b)]$$

$$\int_a^b f(x)dx = S(a, \frac{a+b}{2}) + S(\frac{a+b}{2}, b) - \frac{1}{16}\cdot(\frac{h^5}{90})\cdot f^{(4)}(\xi)$$

Integrare Gaussiană

$$\int_a^b f(x)dx \approx \sum_{i=1}^{m} c_i \cdot f(x_i) \rightarrow \text{grad max de realizabilitate } 2m-1$$

(pt polin de grad max $2m-1$ for- ~~mulele~~ - mulele Gaussiene sunt exacte)

Pt polin Legendre ($P(x)$)

$$c_i = \int_{-1}^{1} \prod_{\substack{j=1 \\ j\neq i}}^{m} \frac{x-x_j}{x_i-x_j}\,dx$$

$$\int_{-1}^{1} P(x)dx = \sum_{i=1}^{m} c_i \cdot P(x_i).$$

Dacă intervalul $\in [a,b]$

$[a,b] \rightarrow [-1,1]$, $t = \frac{2x-a-b}{b-a}$

(sch de var.)

$$\int_a^b f(x)dx = \int_{-1}^{1} f\left(\frac{(b-a)\cdot t + b + a}{2}\right)\cdot\left(\frac{b-a}{2}\right)dt$$

$$\frac{b-a}{2}\cdot\int_{-1}^{1} f\left(\frac{b-a}{2}\cdot t + \frac{a+b}{2}\right)dt$$

$$P_k(x) = \frac{(x-x_1)\cdot(x-x_2)\cdot\ldots\cdot(x-x_{k-1})\cdot(x-x_{k+1})\cdot\ldots\cdot(x-x_n)}{(x_k-x_1)\cdot(x_k-x_2)\cdot\ldots\cdot(x_k-x_{k-1})\cdot(x_k-x_{k+1})\cdot\ldots\cdot(x_k-x_n)}$$

Integrare numerică

$$\int_a^b f(x)dx \approx \sum_{i=1}^{m} c_i \cdot f(x_i)$$

Ex: Presp că vrem x_1, x_2, c_1, c_2.

~~$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$~~

$$\begin{cases} \int_{-1}^{1} 1\,dx = c_1 \cdot 1 + c_2 \cdot 1 \\ \int_{-1}^{1} x\,dx = c_1 \cdot x_1 + c_2 \cdot x_2 \\ \int_{-1}^{1} x^2\,dx = c_1 \cdot x_1^2 + c_2 \cdot x_2^2 \\ \int_{-1}^{1} x^3\,dx = c_1 \cdot x_1^3 + c_3 \cdot x_2^3 \end{cases}$$

$x_1 \neq x_2$

Se ia

$g(x) = (x - x_1)(x - x_2)$

$= ax^2 - bx + c.$

Ec dif cu cond inițiale

Cond Lipschitz, $\exists L > 0$ a.î. $|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$

$D \subset \mathbb{R}^2$ convex $\forall \Leftrightarrow \forall (t_1, y_1), (t_2, y_2)$ dacă tragem o dreaptă între pct, niciun punct nu iese din domeniu

Fie $f(t, y) : D \to$

$\exists L > 0$ a.î. $\left|\frac{\partial f}{\partial y}(t, y)\right| \leq L \quad \forall (t, y) \in D$, at f satisf cond Lipschitz

Dacă f satisf o cond Lipschitz pe D în y at ec. dif cu cond init $y'(t) = f(t, y)$, $a \leq t \leq b$, $y(a) = \alpha$ are o sol unică $y(t)$ pt $a \leq t \leq b$.

O prob. $\frac{dy}{dt} = f(t,y)$, $a \le t \le b$, $y(a) = \alpha$ este bn pusă dacă: $\to \exists$ o sol unică $y(t)$

$\to \exists\ \varepsilon > 0, h > 0$ a.î. $\forall \varepsilon > 0, \forall |\delta(t)| < \varepsilon$ cu $t \in [a,b]$, $|\delta_0| < \varepsilon$ $\frac{dz}{dt} = f(t,z) + \delta(t)$

$a \le t \le b$, $z(a) = \alpha + \delta_0$.

are sol unică ce satisf $|z(t) - y(t)| < k \cdot \varepsilon$

Dacă

Dacă f satisf o cond Lipschitz at ec dif cu cond init, este bine pusă

Met Euler.

$\frac{dy}{dt} = f(t,y)$, $a \le t \le b$, $y(a) \alpha$.

Fie $t_i = a + ih$ pct echidist

$y(t_{i+1}) = y(t_i) + (t_{i+1} - t_i) \cdot y'(t_i) + \frac{(t_{i+1} - t_i)^2}{2} \cdot y''(\xi_i)$

$y(t_{i+1}) = y(t_i) + h \cdot y'(t_i) + \frac{h^2}{2} \cdot y''(\xi_i)$

$y'(t_i)$ sol a ec. dif init $\Rightarrow y(t_{i+1}) = y(t_i) + h \cdot f(t_i, y(t_i)) + \frac{h^2}{2} y''(\xi_i)$, ξ_i

$$\begin{cases} w_0 = \alpha \\ w_{i+1} = w_i + h \cdot f(t_i, w_i) \end{cases}$$

Metode Taylor (Euler Taylor = 1)

$$y(t_{i+1}) = y(t_i) + h\cdot y'(t_i) + \frac{h^2}{2}\cdot y''(t_i) + \dots \frac{h^n}{n!}\cdot y^{(n)}(t_i) + \frac{h^{n+1}}{(n+1)!}\cdot y^{(n+1)}(\xi_i)$$

$$y^{(k)}(t) = f^{(k-1)}(t, y(t))$$

$$y(t_{i+1}) = y(t_i) + h\cdot f(t_i, y(t_i)) + \frac{h^2}{2}\cdot f'(t_i, y(t_i)) + \dots + \frac{h^n}{n!}\cdot f^{(n-1)}(t_i, y(t_i)) + \frac{h^{n+1}}{(n+1)!}\cdot f^{(n)}(\xi_i, y(\xi_i))$$

$\Rightarrow w_0 = \alpha$

$$w_{i+1} = w_i + h\cdot T^{(n)}(t_i, w_i), \quad i = 0, \dots, n-1.$$

$$T^{(n)}(t_i, w_i) = f(t_i, w_i) + \frac{h}{2}\cdot f'(t_i, w_i) + \dots + \frac{h^{n-1}}{n!}\cdot f^{(n-1)}(t_i, w_i)$$

Runge-Kutta

→ vezi la seminar

Met multipas Adams-Bashforth → explicite

Doi pași

$w_0 = \alpha, \ w_1 = \alpha_1$

$$w_{i+1} = w_i + \frac{h}{2}\left[3f(t_i, w_i) - f(t_{i-1}, w_{i-1})\right]$$

Trei pași

$w_0 = \alpha, \ w_1 = \alpha_1, \ w_2 = \alpha_2$

$$w_{i+1} = w_i + \frac{h}{12}\left[23\cdot f(t_i, w_i) - 16 f(t_{i-1}, w_{i-1}) + 5 f(t_{i-2}, w_{i-2})\right]$$

↳ pt a aproxima un pas de timp cunt doar de valoarile pașilor anteriori.

$\Rightarrow \lambda_1 = \frac{6+2}{\cdot} = \frac{8}{\cdot} = 4$

Metode Adams-Moulton. → implicită

$w_0 = \alpha_1, \; w_1 = \alpha_1.$

$$w_{i+1} = w_i + \frac{h}{15}\left[5f(t_{i+1}, w_{i+1}) + 8f(t_i, w_i) - f(t_{i-1}, w_{i-1})\right]$$

↳ aproxfață la pasul curent în fața de valoarea de la pasul curent

$P_1 \rightarrow P_{12}$
$P_2 \rightarrow P_{12} \rightarrow P_{123} = \frac{P_{12}(x - x_3) + P_{23}(x_1 - x)}{x_1 - x_3}$
$P_3 \rightarrow P_{23}$

$\prod (x - x_j)$

↳ nu depinde doar de pașii anteriori

Bonus 2

Se dă regula de mișcare a unui bolid pe circuit $f(x) = \frac{5x+2}{3x+4}$. Dorim să îi aflăm derivata în punctele $x = 0$ și $x = 3$.

Folosim formula în 3 puncte

Fie h distanța dintre puncte, atunci

$$f'(x_0) \approx \frac{1}{2h} \cdot \{ f(x_0+h) - f(x_0-h) \}$$

Alegem $h = 0.1$.

$$f(0+0.1) = \frac{0.5+2}{0.3+4} = \frac{2.5}{4.3}$$

$$f(-0.1) = \frac{1.5}{3.7}$$

$$\Rightarrow f'(0) = \frac{1}{0.2} \cdot \left\{ \frac{25}{43} - \frac{15}{37} \right\}$$

$$f'(0) = 5 \cdot \left\{ \frac{25}{43} - \frac{15}{37} \right\}.$$

$$f(3+0.1) = \frac{15.5+2}{9.3+4} = \frac{17.5}{13.3} = \frac{175}{133}$$

$$f(2.9) = \frac{14.5+2}{8.7+4} = \frac{16.5}{12.7} = \frac{165}{127}$$

$$f'(3) = 5 \cdot \left\{ \frac{175}{133} - \frac{165}{127} \right\}.$$

Bonus

1 Un motor, funcționează după o lege de tipul $f(t) = t^2 + 2t + 7$ unde t = timp. Se cere aflarea lucrului mecanic efectuat de motor în intervalul $[0, 5]$.

Rezolvare:

Alegem ~~10~~ puncte echidistante. (n par)

$x = (0, 0.4, 0.8, 1.2, 1.6, 2, 2.4, 2.8, 3.2, 3.6, 4)$

$y = (1, 1.96, 3.24, 4.84, 6.76, 9, 11.56,)$

$x = (x_0 \; x_1 \; x_2 \; x_3 \; x_4 \; x_5)$

$x = (0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5)$

$y = (1 \quad 4 \quad 9 \quad 16 \quad 25 \quad 36)$

$0,16 + 0,8 =$

$0,64 + 1,6 = 2,24$

$1,44 + 2,4 = 3,84$

1.6

$2.56 + 3.2 = 5.76$

$4 + 4 + 1 = 9$

$24 \cdot 24 = 96$

$5.76 + 5.8 = 11.56$

48

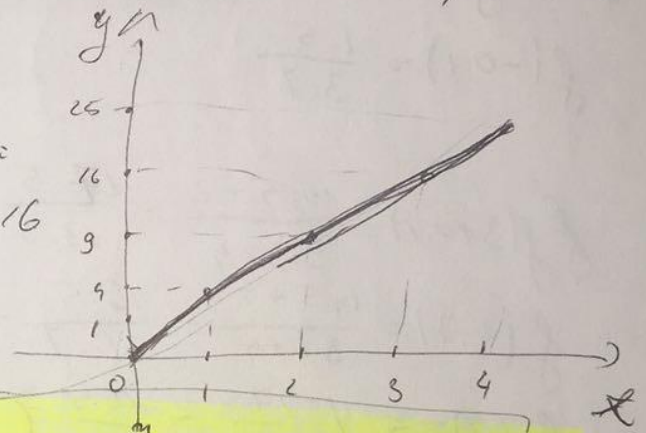
$5.76 + 4.8 + 1 =$

$1 + 2 + 1 =$

$4 + 4 + 1 =$

$9 + 6 + 1 = 16$

$16 + 8 + 1 =$



$$\int_0^5 f(x)\,dx \approx \frac{h}{3} \cdot \left(f(x_0) + 2 \cdot \sum_{j=1}^{\frac{n}{2}-1} f(x_{2j}) + 4 \cdot \sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) + f(x_n) \right)$$

Simpson

$h = 1.$

$f(x_0) = 1$

$\sum_{j=1}^{\frac{n}{2}-1} f(x_{2j}) = f(x_2) + f(x_4) = 34.$

$\frac{n}{2} - 1 = 2.$

$\sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) = f(x_1) + f(x_3) = 4 + 16 = 20$

$f(x_n) =$

CURS-MN 13
+ce ne dă la examen

1) De rez. $y' = y - t^2 + 1$, $0 \le t \le 2$, $y(0) = 0.5$; $y(1) = ?$

Pt examen

Curs 13

- -> cond. L
- -> probl. bine pusă -> sol unică
- -> Forma Taylor
- -> Euler - > orice.
- -> Runge-Kutta -> ideea din spate.
  - -> de recunoscut
  - -> de știut de aplicat
- -> ce e aia multipas -> formă implicită
  - -> explicită

Curs 12

- -> Extrapolare Richardson
- -> complexitatea
- -> ce cuadratură se aplică, de ce se poate aplica, Romberg
- -> ce înseamnă cuadratură adaptivă
- -> să știu de exemplificat (de ex cu Simpson)
- -> integrare Gaussiană, cum aflu gradul de exactitate la $2n-1$
- -> cum putem face fără a rez sist cu polin Legendre.

C11

- -> ce înseamnă să integrăm/derivăm un set de pct ✓
- -> polinomul Lagrange, formule ✓
- -> derivare numerică cu Taylor la o formulă la care știm și multe
- -> ce înseamnă cuadratură numerică
- -> Regula Simpson -> pt ele se fol Taylor. ✓
- -> trecerea la formule compuse -> posibil să ne dea ex cu așa ceva ✓
- -> Newton-Cotes
- -> ~~Formule~~ dif de la compusă Simpson la compus la-trapez.

C10

- -> aprox în sensul CMMP -> unde se practică ✓
- -> de unde se scot coef? Cum minimizez eroarea? ✓
- -> discuțiile cu produs scalare
- -> cum arată sist
- -> ce înseamnă sist GRAM, coef Fourier, optimizare pt a scăpa de sist

1

-> Se sparge sol în două
-> Să știm să demonstrăm că ambele sunt base ort.
-> Cum se consideră prod. scalar.
Fct raționale -> aprox cu $r(x) = \frac{p(x)}{q(x)}$, cerea cu semn Mclorin
Ce înseamnă FFT., cum se face, ce face ... (cu nr complexe)

COD ✓

-> Curs de interpolări -> TOT ✓
-> Nu ne dă dincuț de la DS Derivate ✓
-> Dar ne cere formele finale și conceptul din spate ✓
-> Trecerea către ele.
-> Formulele de interpolare Newton
-> Ce înseamnă combinările ✓
-> Ce înseamnă cond Hermite
-> Trecerea de la Lagrange la Spline, de ce apar erorile la Lagrang
-> Splini -> poate să dea orice :)), de unde știe, cond, ce înseamnă cond suport, cum arată matricile ✓
-> Polin Bernstein + Curbe Bezier (cum arată, proprietăți, construcție cu Castel-Bejou) leg cu polin Bernstein)

08 ✓

-> curs de matrici -> alg QR -> tol (fără matr. nesimetrice)
-> ce înseamnă SVS ✓
-> Construcție USV^T +0.1
$alg\ S = (A \cdot A^T)$